Report for STATA Lab Final

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1 Introduction

A vast and evolving body of literature spanning over a century has been dedicated to studying and predicting revolutions, their causes, their outcomes, and the determinants to their occurrences and results. The central question that motivates this report is an extension of the task of predicting revolution, and it asks, "Can the effect of the determinants of revolutionary outcomes be *measured*", or that how much does a change in each determinant affect the probability of success of a revolution.

2 Data Description

We begin by summarizing the two binary variables, since they have the fewest dimensions and thus are the easiest to summarize. As seen in Table 1, the sample contains 40 revolutionary attempts, a very small sample size given the rarity of revolutionary events. The distribution of success/failure outcomes, as well as the widespread of the revolts is fairly equal (roughly 50% in both cases), with no skews towards either extreme.

Table 1: **Summary of revolutionary outcomes and the widespread of the revolts in the sample**

Immediate Outcome	Frequency	Percent	Is the revolt widespread?	Frequency	Percent
Failure	21	52.50	Yes	21	52.50
Success	19	47.50	No	19	47.50
Total	40	100.00	Total	40	100.00

Source: Goldstone et al. (2022), recent events compiled by the author. Widespread of the revolts compiled by the author

Below is a summary of the remaining non-binary statistics of the sample. The indices are taken from the Fragile State Index database; the higher an indicator is, the worse it is to the stability of a state. In particular, a higher State Apparatus Index means that the state in question faces more threat from criminal organizations, terrorism, coups, etc., and the security apparatus is also more willing to crack down on perceived enemies. A higher Factionalized Elite means the country faces serious fragmentation along identity lines such as races, ethnicities, or religions, as well as more serious gridlock and struggle among the rulling class. A higher Public Service Index indicates a lower ability of the state to provide public service and safety to the public, and/or such service is only provided to the rulling class. A higher External Intervention Index indicates a higher chance of foreign intervention in a state, especially in time of crisis.

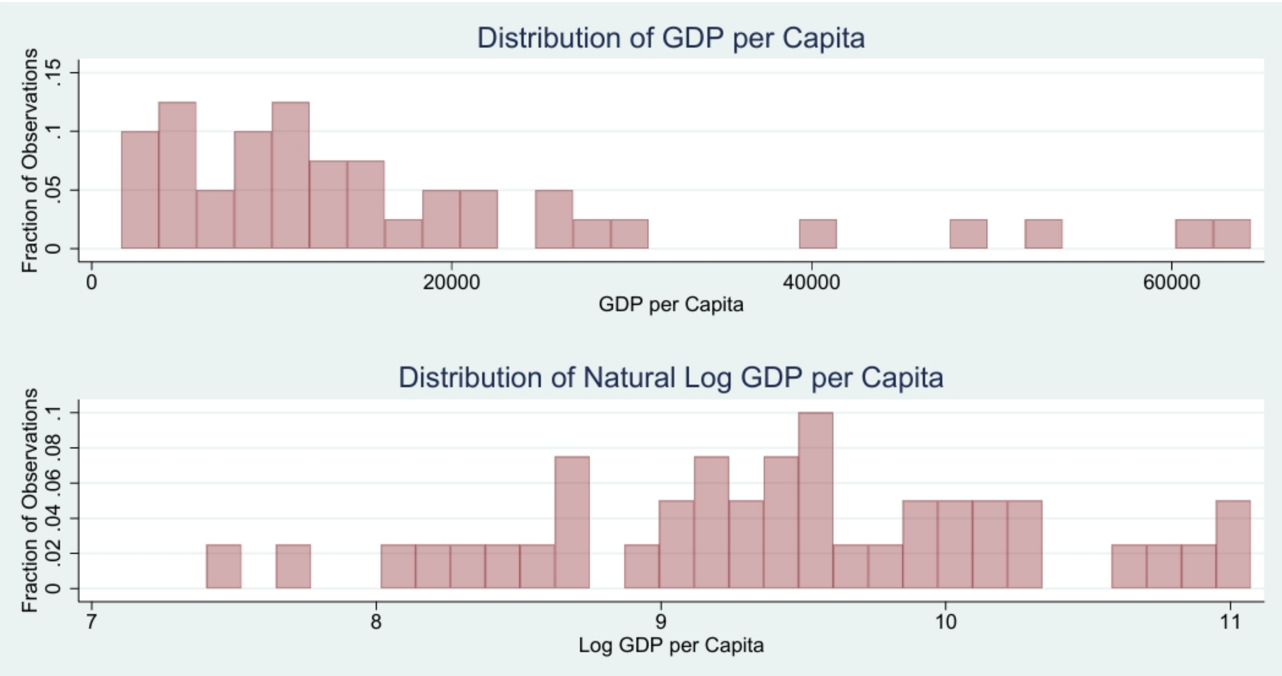
Table 2: **Summary of non-binary statistics in the sample**

Category	Number of obs	Mean	50th percentile	Std. dev.	Min.	Max.
GDP per capita	40	17,604.09	12,183.00	15,849.25	1,642.90	64,401.51
Factionalized Elite Index	40	7.77	7.90	1.57	2.20	9.70
Security Apparatus Index	40	6.39	6.50	1.65	2.10	9.00
Public Service Index	40	5.66	5.60	2.01	1.20	8.90
External Intervention Index	40	5.87	5.80	1.93	1.30	9.70

Source: Data for GDP per capita is from the World Bank, or the CIA Factbook where World Bank data is not available. The indices are taken from the Fragile State Index dataset. Data range is from 2007 to 2024. GDP Per capita is in current international $

These data could be further visualized by graph. In the case of GDP per capita, the data is heavily skewed left, which justifies the use of natural logarithm. Figure 1 shows that natural logarithm of GDP per capita has a more normal distribution.

Figure 1: Comparison of distribution of raw GDP per capita and its natural logarithm

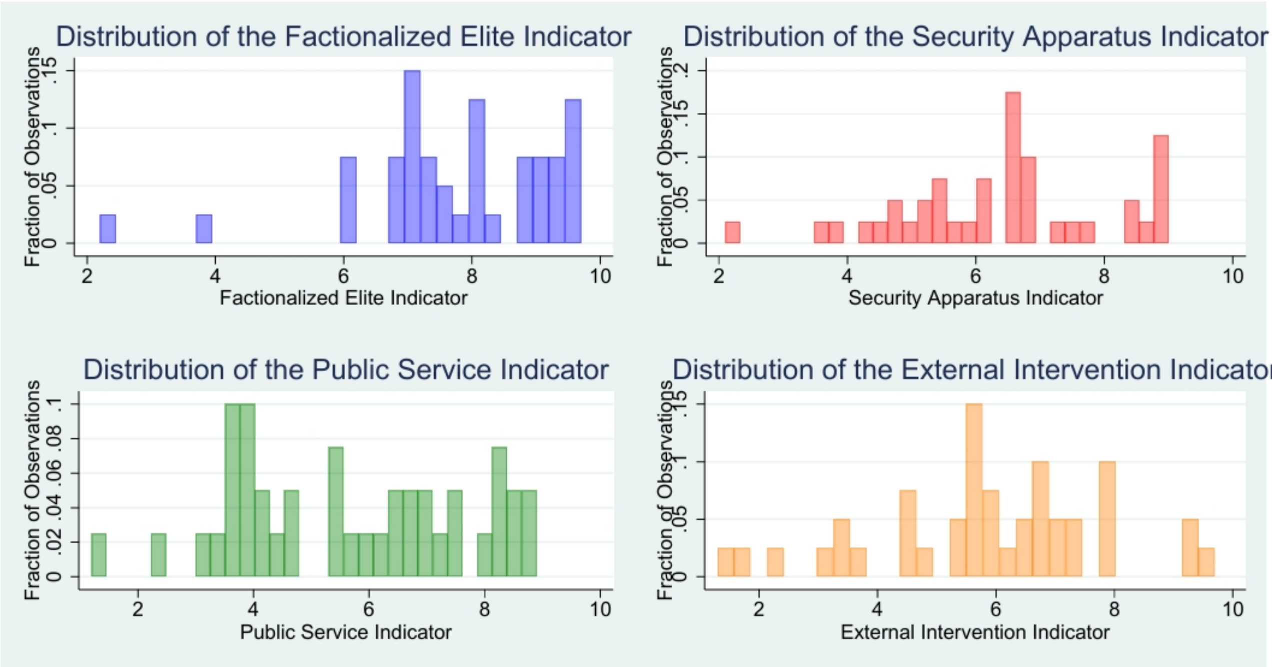


Source: The World Bank and the CIA Factbook. Range: 2007 - 2024.

By contrast, the Fragile State indicators mostly have a normal distribution, with the exception of the Factionalized Elite index, which shows skewness toward the higher extreme. Coupled with the information from Table 2, we can infer that on average, the sampled countries exhibit high level of factionalization among identity lines, and/or high level of gridlock and struggle among the ruling class. The other indicators also fall in the range of low to medium warning level on average[1].

[1] According to the Fragile State Index, a total score of 60-69.9 is considered low warning, while 70-79.9 is warning. Divide the score among 12 indicators, and we have a low warning score of 5-5.83 per indicator as low warning, and 5.83 - 6.67 as warning.

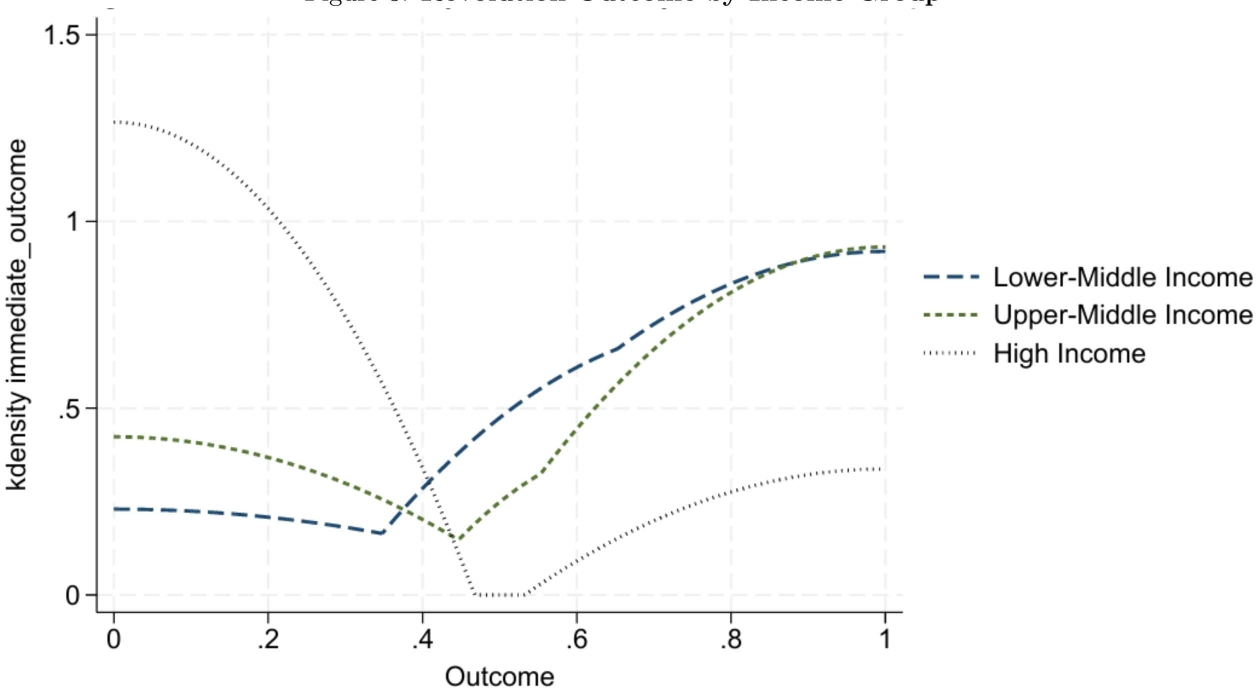
Figure 2: **Comparison of distribution of the Fragile State Indicators**



Source: Fragile State Index. Data range: 2007 - 2024

We also compare revolution outcome by income groups in Figure 3. Countries that see the fewest instances of success are countries with higher income, while better outcomes are seen in countries with lower incomes. There are no low-income countries in the sample.

Figure 3: **Revolution Outcome by Income Group**

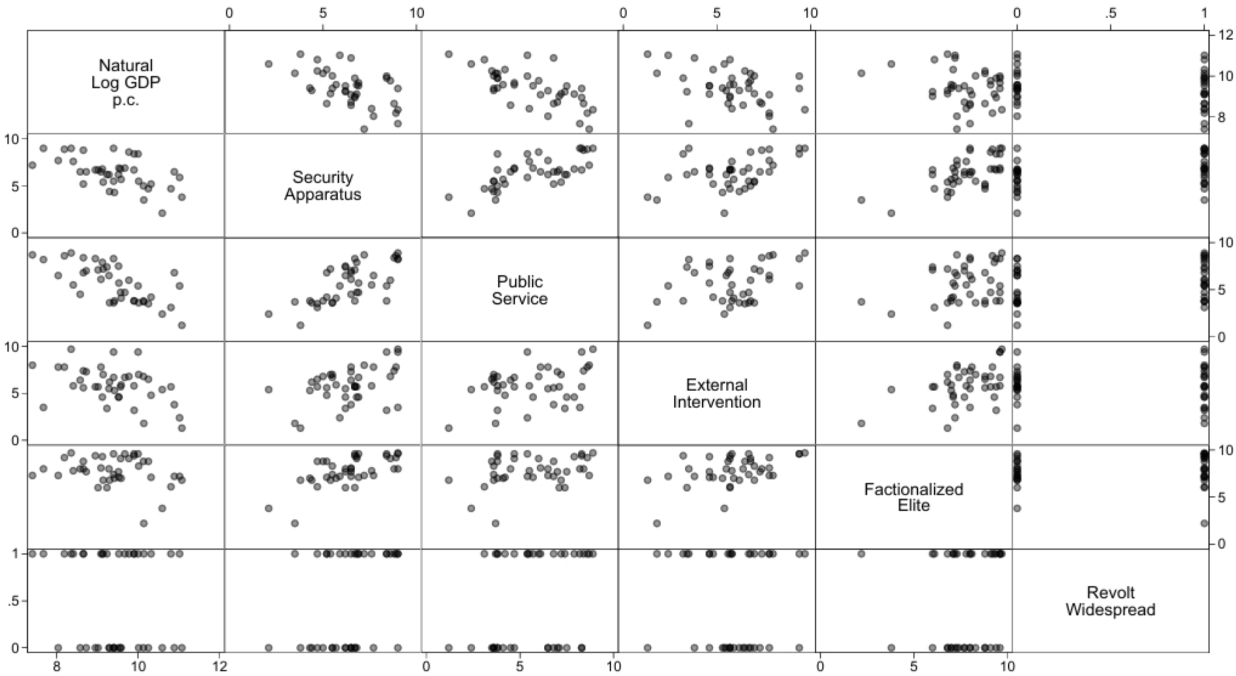


3 Regression

3.1 Pre-Regression Check

It is useful before regression to check for any signs of multicollinearity. As seen in Figure 3, identifiable patterns can be seen between natural logarithm of GDP per capita and the indicator for Security Apparatus, as well as that for Public Service, though the correlations are not visibly significant. More concerning is the relationship between Security Apparatus and Public Service, which follows an identifiable and positive pattern: a stronger security apparatus means a weaker public service (higher score). However, overall, based on the graph, multicollinearity between variables does not seem to be a major concern.

Figure 4: **Scatterplot matrix showing correlation between independent variables in the study**



3.2 Regression Result

For this model, we use Firth logistic regression to account for the small sample size. We first run the model with all variables, and obtain the following:

Table 3: **Full Model: Impact of different variables on revolutionary outcome**

Variable	Coefficient	SE
GDP per capita (expressed as ln)	-1.88** (0.017)	0.79
Factionalized Elite Index	-0.14 (0.641)	0.31
Security Apparatus Index	-0.47 (0.293)	0.45
External Intervention Index	0.24 (0.346)	0.26
Public Service Index	-0.13 (0.692)	0.32
Is the revolt widespread?	-0.21 (0.795)	0.81
Number of observations	40	
Chi-squared	7.79	
p-value	0.2536	
Penalized log likelihood	-12.61	

Note: *p-value of each variable in parentheses.*
** p<0.1, ** p<0.05, *** p<0.01*

As we can see, with the exception of GDP per capita, the remaining variables are very statistically insignificant. Moreover, the model itself is unusable. We tweak the model by removing variables that are the least significant, until we get:

Table 4: **Model with reduced number of variables: Impact of different variables on revolutionary outcome**

Variable	Coefficient	SE
GDP per capita (expressed as ln)	-2.05*** (0.004)	0.72
Security Apparatus Index	-0.64*** (0.041)	0.31
Number of observations	40	
Chi-squared	8.17	
p-value	0.0169	
Penalized log likelihood	-17.12	

Note: *p-value of each variable in parentheses.*
** p<0.1, ** p<0.05, *** p<0.01*

3.3 Post-Regression Diagnostic Test

Logistic regression model requires a special treatment to get Pearson residuals. Furthermore, as seen in the figure below, the residual of this model does not follow the assumption of homoskedasticity, nor does it strictly follow a normal distribution. This is normal in a logistic model: residuals are not assumed to be homoscedastic or normal.

Figure 5: **Pearson residual plotted against Predicted probability**

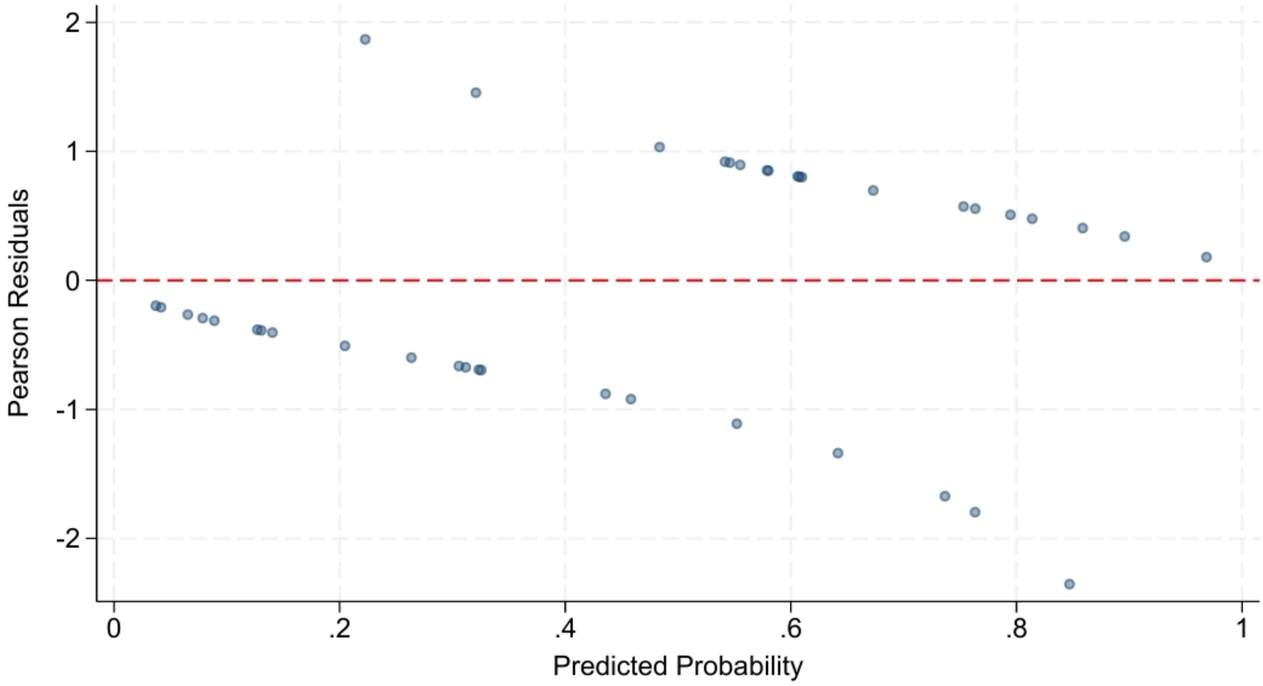
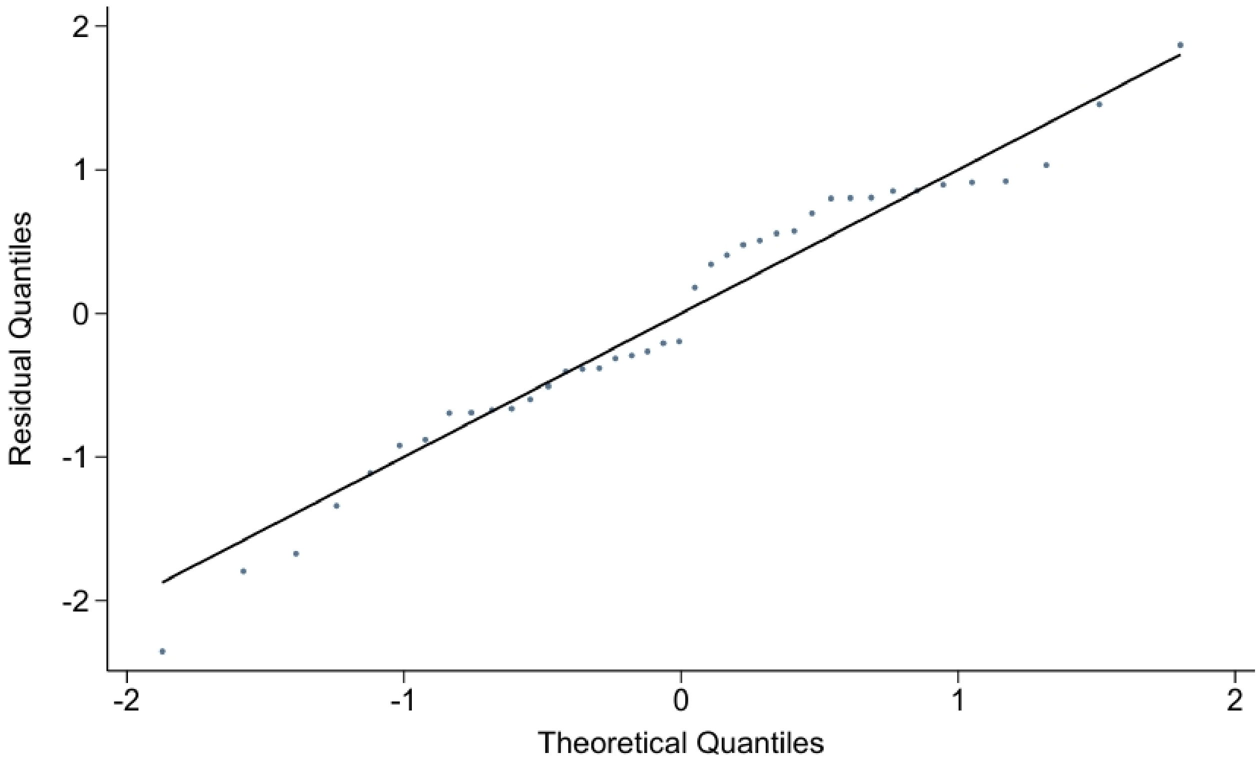


Figure 6: **Normal quantile plot of residual**



4 Discussions

According to the final model, GDP per capita and Security Apparatus Index are inversely related to the probability of a revolution being successful. An increase of one unit in GDP per capita (expressed in natural logarithm) is associated with a decrease of 2.05 unit in the log-odd, keeping the other variable the same. With the same context, an increase of one unit of Security Apparatus Index is associated with a decrease of 0.64 unit in the log-odd. These results are consistent with literature: Shugart (1989) suggests that a semi-modern state with a large number of urban and middle-population will be able to better resist a revolt. The Security Apparatus Index is associated with a state's willingness to crack down on revolts: the higher the score, the more willing the state to use extreme violence. A state more willing and capable of suppressing revolts would reduce the probability of a revolution being successful, as also suggested by literature. This was also suggested by Shugart (1989), and was already alluded to by the first literature on revolutions. An obvious policy suggestion is that state focusing more on economic development with all of its necessary reforms, as the GDP per capita term is more strongly associated with reducing revolution success than the security terms. A cynical path that a state can, and has taken, is to focus solely on increasing its security to clamp down on any revolutionary attempts. Given that a more violent and secure security apparatus is negatively correlated with the economy, this path can have mixed outcomes.

In terms of prediction, the model does a good job at predicting revolutions, though given the small sample size, one can argue that this is a result of overfitting. Still, this is a result worth paying attention to.

Table 5: **Predicted Outcomes and Actual Outcomes**

Actual Outcome	Predited Success	Predicted Failure
Success	16 84.21%	3 15.79%
Failure	5 23.81%	16 76.19%

One limitation of this research is that the sample size is very small, at 40 observations. Future research can expand the sample size by including countries where revolution *did not happen* within the same time period for larger sample size and a more comprehensive model. Secondly, one can criticize the use of the Fragile State Indicators in this report, as the methodology of the index itself has been subject to criticism and scrutiny over the years, and as the scope of each indicator are larger than that demanded by each variable.

Thirdly, one can argue that a regression model (at least a model expressed such as the one in this report) is not the right way to understand revolutions. Revolutions are social phenomena with complex causes, depending on the context special in each individual country, as well as the way each determinant interact with each other. The point of regression models is to *isolate* the marginal effect of each variable, keeping the other variable constant. This assumption might not be true in this case, as, for example, if one is to increase GDP per capita, one will have to enact reforms that might reduce the strength of the security apparatus, and the reaction of the security apparatus might impact economic growth and by extension the chance of revolution. Future research might take into account these interactions for a more complete model.